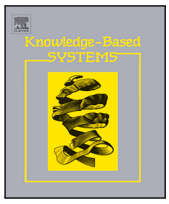
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@LM DeceptionNet: A multimodal approach for efficient transfer learning-based deception detection

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ABSTRACT

In terms of deception detection, traditional contact-based techniques often require collecting physiological signals, which can negatively impact device accuracy and participant comfort. While multimodal features extracted from audio and video modalities have been shown to outperform human observers on public datasets, the generalizability of existing audio and visual-based deception detection methods in different scenarios remains insufficiently explored. To narrow this gap, this work proposes a novel domain knowledge transfer learning method for deception detection in cross-scenario applications, which enhances its generalization and adaptability. Additionally, we designed a multimodal framework that filters out irrelevant information from other modalities when a particular modality yields reliable results, further improving overall system accuracy and robustness. We evaluate the proposed method on different public datasets, achieving promising generalizability results with consistent enhancements using four variations and networks. Apart from this, the proposed @LM DeceptionNet demonstrates better generalization capacity in computational efficiency, feature extraction, and adaptability compared to a larger model when employing fewer parameters.

1. Introduction

Deception detection is a complex interdisciplinary field that has garnered extensive attention, and most applications are employed in criminology, law enforcement, border security, business, and healthcare [1–10]. To enhance the accuracy and efficiency of deception decision-making processes, we aim to strengthen security and ethics and prevent significant losses due to deceptive activities [11]. Most recently, deep learning techniques have proven indispensable in deception detection. For instance, deep learning can capture various features in multiple modalities, leading to better performance in deception detection [12–14]. Despite these advancements, previous works under various dataset scenarios suffer from annotated bias in unimodal features such as visual, gaze, speech, audio, and Electroencephalogram (EEG), which leads to the adoption of unimodal features in deception detection. Additionally, deception detection approaches for audio and video remain largely unexplored in terms of their generalizability [15].

Multimodal deception detection methods demonstrate appealing performance and promising results in deception detection tasks. Current detection models explore a variety of unimodal and multimodal detection methods, e.g., audio, speech, visual, speech-visual, gaze-EEG, EEG-audio, and audio-visual frameworks. Among these, audio-visual-based models [2,12,15,16] detect deceptive behaviors using audio and facial movements. The advent of deep learning has seen a variety of techniques and algorithms used to achieve state-of-the-art results. However, these methods are limited to detecting deception in a single scenario and are not able to generalize for multiple deception scenarios. In fact, the generalizability of existing methods in different scenarios is relatively unexplored, which is attributed to two factors. First, deceptive models in a single scenario are inadequate to accurately represent all other scenarios. As such, deception detection approaches require cross-scenario capacity. Second, since cross-scenario models need to pair multimodal features, they usually require more natural datasets with relatively multiple features compared to lab-based datasets.

To address the aforementioned issues, we propose a novel approach to effectively transfer knowledge from deception detection models trained on courtroom trial data to game show scenarios, named @LM DeceptionNet, which achieved good performance in cross-scenario applications. Specifically, this approach applies transfer learning, as illustrated in Fig. 1, to optimize the model's adaptability in different scenarios [17]. Our approach involves transferring a pre-trained model from the Real Life Trials dataset while retaining its weights to preserve foundational features. To improve adaptability, we implement a two-phase fine-tuning process: initially freezing the parameters of the pre-trained model and training only the newly added layers, followed

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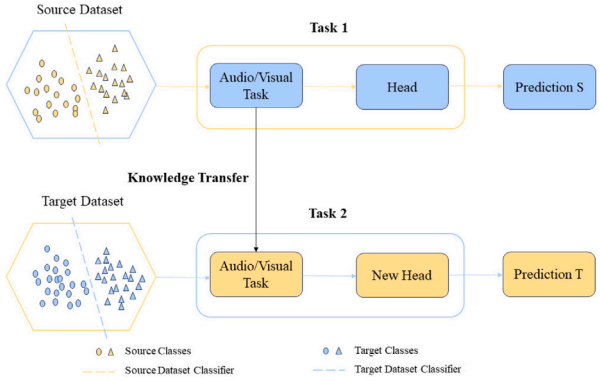


Fig. 1. General illustration of transfer learning processing. First, train a source model on the source dataset to ensure it performs well on that task. Next, apply the source model to the target dataset and fine-tune it. Afterward, evaluate the performance on the target task and optimize accordingly.

by selectively unfreezing specific layers to fine-tune them on the Box of Lies dataset and DOLOS dataset. This design enables the model to learn features relevant to the new scenario and generalize well in diverse deception detection contexts. Additionally, we utilize a late fusion strategy that combines decisions from each modality using a label prediction rule, enabling cohesive and adaptive multimodal fusion. Our main **contributions** are summarized as follows:

1. We propose @LM DeceptionNet, a transfer learning-based framework for cross-scenario deception detection. Our @LM DeceptionNet marks the first attempt to simultaneously cover multimodality and generalizability in deception detection methods. We exploit three scenario sampling strategies i.e., scenario simultaneous, scenario alternating, and scenario-by-scenario—to conduct cross-scenario testing. This design effectively utilizes learned features in new domains, significantly boosting the model's generalizability and efficiency.
2. We employ dual Transformers with a novel attention mechanism to boost classification accuracy while reducing computational complexity. This design emphasizes the necessity of specialized visual models for deception detection tasks. Simultaneously, the introduced attention mechanism balances computational efficiency with the need for comprehensive context modeling, facilitating high-accuracy detection in diverse and complex environments.
3. We develop a multimodal fusion method integrating audio and visual features for deception prediction. To integrate predictions from multiple independent models, our fusion strategy reduces errors from single modalities and enhances overall decision accuracy and reliability. This approach effectively addresses the multimodal fusion problem using deep learning techniques for efficient multimodal data processing.

The rest of this paper is organized as follows. Section 2 reviews existing research in deception detection. Section 3 illustrates the methods and framework we propose. Section 4 details the experimental setup and the metrics used for evaluation. Section 5 discusses the analysis of our findings. Section 6 describes the applications of deception detection. Section 7 concludes this paper and with future work.

2. Related works

2.1. Deception detection datasets

Earlier deception detection datasets focused on the psychological underpinnings of deception [25–31], identifying potential indicators such as physiological responses, visual signs, vocal patterns, verbal content, and behavioral features [32]. Among the most significant advances in the field was the development of the Real-Life Trials (RLT) dataset [18], which draws on courtroom proceedings. This scenario offered natural instances of deception, and recent papers that achieved nearly perfect results utilized this dataset. To amplify the resources available, Pérez-Rosas et al. [19] and Kamboj et al. [20] introduced two new datasets collected from street interviews and political speeches. Despite these contributions, verifying the authenticity of participants' responses in these datasets remains challenging. To address these drawbacks, fewer researchers developed laboratory-based multimodal deception detection datasets. These datasets involve scenarios where participants are permitted to exhibit deceptive behaviors, such as describing objects [21], discussing personal characteristics [22], or engaging in face-to-face interviews [33]. Nonetheless, the controlled environment may result in low motivation to deceive, leading to fewer detectable deceptive cues. Subsequently, a gameshow-based deception detection dataset was presented in Box of Lies [24], which primarily included verbal and visual features. In contrast, the DOLOS dataset offers a rich set of annotations in multiple modalities, including audio and visual data. In essence, to find the same multiple modalities in different datasets for transfer learning, we evaluated the above datasets in Table 1. As a result, there are transfer-learnable and effectively aligned data modalities in the Real Life Trials (RLT) dataset, Box of Lies dataset and DOLOS dataset, which can be employed in our proposed deception network.

2.2. Multimodal deception detection

Multimodality learning is an approach in machine learning that focuses on integrating information from multiple sensory or data modalities. This technique contributes to creating models that can understand and explain data from varied sources such as text, images, audio, and more, effectively combining them into an overarching representation. The goal is to exploit the strengths of each modality to enhance the overall performance and accuracy of the model. As samples from different modalities are mapped into a common semantic space, multimodality learning enables the measurement and comparison of semantic similarities between them. To achieve this goal, many previous approaches have been posited in the last decade [13,14,34–36]. Multimodality feature fusion can be divided into four categories: feature-level fusion, decision-level fusion, hybrid fusion, and model-level fusion. Feature-level fusion (also known as early fusion) is the most common strategy in multimodal recognition systems. This strategy involves concatenating features extracted from different modalities into a single high-dimensional feature vector immediately after extraction [4]. In contrast, late fusion aggregates potential representations from each modality into a single vector for classification [37] or aggregates modality-specific decisions based on voting [38]. Hybrid fusion is designed to aggregate early features while spreading the losses to separate extractors. This fusion method is infrequently used in deception detection because neural networks are not usually preferred in task training. Finally, model-level fusion employs multi-stage recurrent networks to process time-aligned multimodal data as in [39]. However, the sparsity of deception detection data complicates learning from feature perturbations, and complicated fusion techniques can also lead to model overfitting.

To reduce the occurrence of overfitting, recent studies combine verbal and nonverbal features to boost the performance of deception detection and present effective fusion approaches [40–43]. For example, several papers analyzed facial features in RGB images [3,40,42] to predict deception. Other visual features have also been utilized to enhance the accuracy of deception detection [4,43]. Additionally, audio features, including MFCC and raw audio, have been incorporated to boost performance [4,5,40]. Moreover, Karnati et al. [41] explored the EEG signal for deception detection. In terms of deception detection, the

Table 1
Comparison of deception detection datasets.

Dataset	Year	Subject	Lie/Truth ratio	Modality	Scenario
Real Life Trials [18]	2015	56	1.02	audio, text, visual	Court-room
Real Life Deception [19]	2015	–	1.24	text,visual	Street interviews
Political Deception [20]	2020	88	–	audio, text, visual	Political speech
Bag of Lies [21]	2019	35	0.99	audio, EEG, gaze, visual	Lab
MU3D [22]	2019	80	1.00	observer ratings, text, visual	Lab
Deception Detection and remote PPG [23]	2021	70	0.60	audio, physiological, visual	Lab
Box of Lies [24]	2019	26	4.61	text, visual	Gameshows
DOLOS [16]	2023	213	1.16	audio, text, visual	Gameshows

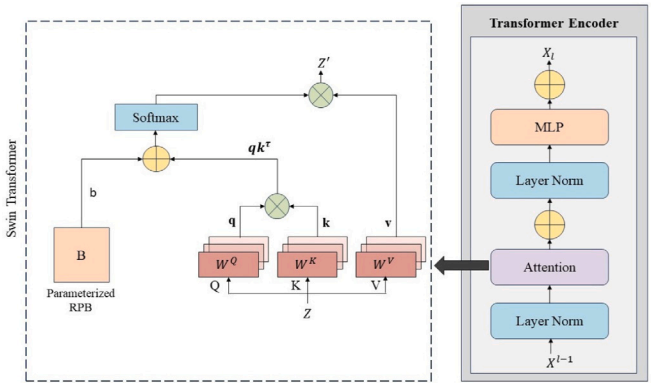


Fig. 2. Swin Transformer. It proposes a hierarchical network structure to address the multi-scale issues of visual images. Additionally, it introduces Shifted Windows, which significantly reduce the computational complexity of the transformer.

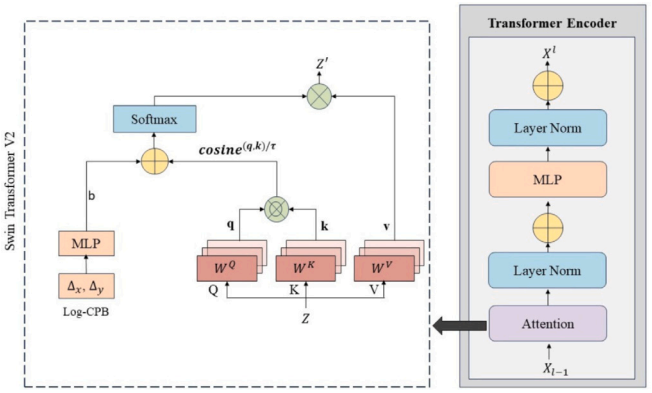


Fig. 3. Swin Transformer V2 tackles three primary issues encountered in the training and application of extensive visual models: instability during training, differences in resolution between pre-training and fine-tuning phases, and the significant need for labeled data.

sparsity and variability of data can complicate learning from feature perturbations, and overly complex fusion techniques may lead to model overfitting. As such, our work builds on these existing studies and proposes a novel multimodal learning framework that combines audio and visual features for deception detection in diverse scenarios. We utilize a decision-level fusion strategy for our deception detection tasks while maximizing the utilization of multimodal data.

2.3. Transformer, attention, agent attention

A recent study demonstrated that the transformer model produced exceptional results in image classification [44–46] and multimodal tasks [47]. Originally deployed in natural language processing, it has now emerged in computer vision. The attention mechanism performs various functions in computer vision and emulates human cognitive processes by identifying relevant input locations, evaluating channel contributions, and capturing temporal dependencies [46,48]. The self-attention mechanism maps relationships between segments and their positions to mold representations of sequences. Self-attention was originally conceived for natural language tasks [49] and was then broadly adopted as part of the transformer architecture [50].

Transformers and self-attention face multifaceted challenges when integrated into the visual domain. Due to the quadratic complexity of the Softmax attention mechanism commonly used in transformer models, which calculates similarity between query-key pairs [51], directly applying this attention to global visual tasks can be computationally prohibitive. In response, recent works have attempted to reduce computational demands through the development of more efficient attention mechanisms. For example, Figs. 2 and 3 illustrate the frameworks of the Swin Transformer [52] and Swin Transformer V2 [53], respectively. The Swin Transformer improves computational efficiency by limiting self-attention computations to smaller windows and reducing the receptive field. Swin Transformer V2 addresses three main challenges associated with the training and deployment of large-scale visual models, compared to the Swin Transformer. The PVT [50] reduced computational load by lowering key and value resolution and implementing sparse global attention. Although these approaches effectively reduced computational complexity through localized attention computations, they limited the model's ability to capture long-term dependencies.

In terms of computational cost, Softmax provides strong expressive power, whereas linear attentional mechanisms deliver simplicity in computation but have limited expressive power. These two types of attention have been regarded as distinct techniques, with some researchers focusing on reducing the computational demands of Softmax attention, while others aim to enhance the efficacy of linear attention. For instance, Dongchen Han [54] introduced a novel attention paradigm called Agent Attention, combining the low computational overhead of linear attention with high expressiveness. Unlike previous works, we employ four variants of our visual model: @LM DeceptionNet-ST, @LM DeceptionNet-STV2, @LM DeceptionNet-AAST, and @LM DeceptionNet-AASTV2. This demonstrates that our proposed approach achieves competitive detection accuracy while also being efficient and comparative in terms of model complexity, parameter count, and inference speed.

2.4. Transfer learning

Given the complex relationships between features and learning outcomes, transferring relevant information about these variables can be beneficial. Even though transfer learning has been explored in a multitude of fields including natural language processing [55,56] and computer vision [56–58], research on using transfer learning to decipher learning outcomes and structures, particularly in the context of deception detection-based learning, remains relatively scarce. The aforementioned research has utilized pre-trained transformers as verbal cues to support transfer learning in verbal-based deception detection [59]. Other research has ascribed multimodal data to learning for deception detection and showed boosted multimodal detection

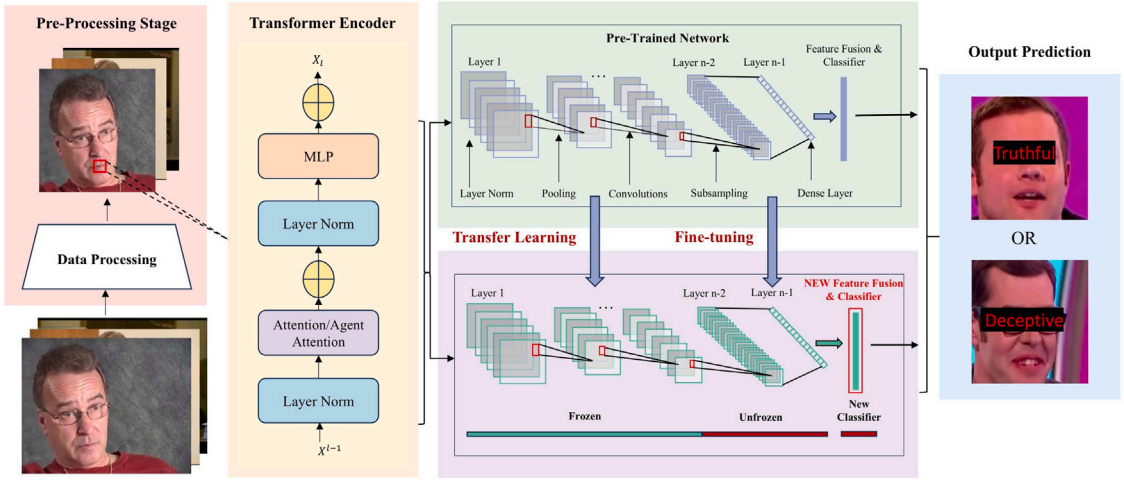


Fig. 4. The overall architecture of @LM DeceptionNet: In the pre-processing stage, input images are resized to ensure consistent dimensions and normalized for subsequent analysis. Subsequently, the resulting feature maps are processed through a transformer encoder employing self-attention or agent attention targeting the sampling points. Following this, the framework employs a pre-trained model, which undergoes fine-tuning in the transfer learning stage. Ultimately, this process culminates in the generation of predictive outputs for detecting deception.

compared to one-modal baselines [3,4,16]. It is possible to refine the model's performance even further by migrating relevant knowledge from other datasets, but such algorithms have not been extensively explored for deception detection. In light of this premise, our approach proposes the application of a comprehensive framework for multimodal data analysis, detection modeling, multimodality learning, and transfer learning.

Transfer learning focuses on enhancing the predictive capabilities of models in the target domain by leveraging knowledge from the source domain [23,60]. This method is particularly significant in the realm of deep learning, where model performance heavily depends on the availability of well-annotated data. Transfer learning can be categorized into three distinct types: unsupervised [60], inductive [61], and transductive [62] transfer learning. Both unsupervised and inductive transfer learning tackle different tasks between the source and target domains. However, inductive transfer learning specifically uses limited labeled data from the target domain to develop a model suited for that domain, while unsupervised transfer learning does not rely on labeled data from either domain during the training phase. Transductive transfer learning, on the other hand, is applied when the tasks across domains are the same but the domains themselves differ.

Recently, numerous advanced techniques in transductive transfer learning have been developed. For example, Ding et al. [63] created a graph-based model that optimizes target labels alongside domain-invariant features. Long et al. [64] developed a residual transfer network that simultaneously trains an adaptive classifier and transferable features. Haitham Bou Ammar [65] integrated lifelong policy gradient reinforcement learning with cross-domain knowledge transfer, enhancing continuous knowledge transfer across domains, particularly suitable for decision-making tasks like Dialogue Policy Learning (DPL) in dialogue systems with scant data. Tzeng et al. [66] outlined an adversarial transfer learning framework introducing the ADDA technique, and Tongliang Liu [67] proposed an analysis method showing that similar feature structures between domains facilitate effective feature transfer, thus stabilizing the learning process. Zhang et al. [68] challenged the traditional assumption that source and target label sets are identical, exploring scenarios where the target label set is a subset of the source label set. Our approach builds on the above research through the introduction of a novel transfer learning framework for multimodal deception detection. This framework exploits pre-trained and fine-tuned models to enhance cross-domain learning. We aim to analyze the Box of Lies, DOLOS and Real Life Trials datasets to expand model adaptability and generalizability in previously unexplored deception detection scenarios. We employ a transductive transfer learning approach to ensure our model performs well in different but related scenarios, maintaining high performance despite scenario shifts. Our work integrates advanced transductive techniques to facilitate efficient transfer, focusing on applying similar feature structures for stable learning in related domains.

However, the cross-scenario and variants-based model are unexplored in deception detection. Therefore, we propose @LM DeceptionNet, which contains four variants in our model and adopts a transfer learning strategy for cross-scenario architectures. An additional benefit of our multimodal deception is that it effectively boosts accuracy rates and reduces computational complexity while minimizing accidental errors.

3. Methodology

To exploit prior knowledge from audio and visual cues in existing deception detection databases, we employ a separate model for each modality and utilize a transfer learning strategy for visual task deception detection. After that, we apply the late fusion strategy in multi-modality deception detection. The overarching framework of our @LM DeceptionNet is depicted in Fig. 4, which includes data processing, the transformer encoder part, pre-trained architecture, transfer knowledge, and the fine-tuning phase. The visual features are processed in four variants of our proposed method, while the CNN module handles the audio input.

3.1. Audio feature learning

Wav2vec 2.0 consists of a CNN-based feature encoder, a transformer-based context network, and a quantization module [69]. The feature encoder comprises seven blocks, each containing a one-dimensional temporal convolution with 512 channels, followed by layer normalization [70] and GELU activation [71]. In this subsection, raw audio data is processed through a 1D-CNN module, resulting in feature representations denoted as $X_a \in \mathbb{R}^{L\times D}$, where L represents the sequence length and D the dimensionality of each sequence embedding. Each audio sample features a dimensionality of 64×512. For audio modality analysis, we employ Wav2vec 2.0 as the primary audio modality network architecture, trained by optimizing the cross-entropy loss L_a [69].

Cross-entropy loss, also known as log loss, is a commonly used loss function in classification problems. It aims to reduce the discrepancy

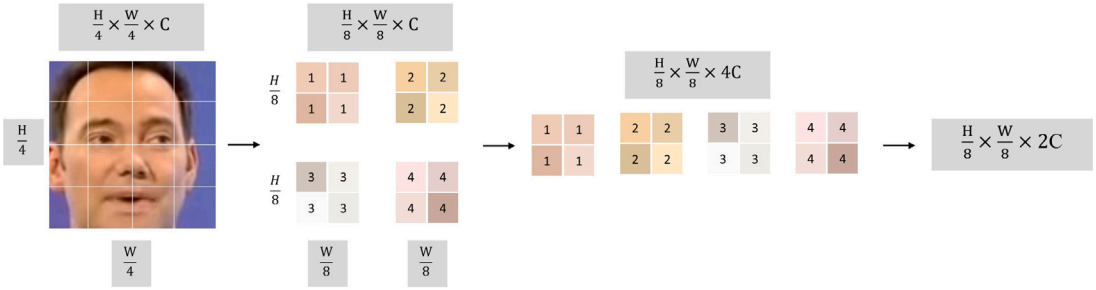


Fig. 5. Swin Transformer procedure of stage 2. This stage contains a Swin Transformer Block and Patch Merging. Patch Merging acts as the inverse operation of pixel shuffle, where the spatial dimensions are halved, and the number of channels will multiply by 2.

between the predicted probability distribution and the actual label distribution. In our context, cross-entropy loss is defined as:

$$L_a = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{1}$$

Here, N represents the sample size. y_i denotes the actual label of the i-th sample, taking values of 0 or 1. $\hat{y}_i$ represents the model's predicted probability that the i-th sample belongs to the positive class.

This form of cross-entropy loss is intended to be minimized through optimization methods such as gradient descent, aligning the model's predictions as closely as possible with the true labels. Hence, our proposed method effectively explores deep learning techniques in the audio processing field, achieving more accurate audio analysis and processing results.

3.2. Vision feature learning

3.2.1. Visual transformer

In this phase, we adopt four variants of our proposed visual model, namely @LM DeceptionNet-Swin Transformer (@LM DeceptionNet-ST), @LM DeceptionNet-Swin Transformer V2 (@LM DeceptionNet-STV2), @LM DeceptionNet-Agent Attention Swin Transformer (@LM DeceptionNet-AAST), and @LM DeceptionNet-Agent Attention Swin Transformer V2 (@LM DeceptionNet-AASTV2). In @LM DeceptionNet-ST, we employ the Swin Transformer as the backbone. The Swin Transformer, a hierarchical network structure based on the transformer, addresses the multi-scale problem in visual images [52]. The introduction of Shifted Windows reduces the computational complexity of the transformer. This complexity can be expressed as:

$$\Omega(\text{MSA}) = 4HWC^2 + 2(HW)^2C, \tag{2}$$

$$\Omega(\text{W-MSA}) = 4HWC^2 + 2M^2HWC, \tag{3}$$

In this backbone, image features $[H, W, C]$ are divided into multiple patches during the Patch Partition stage, with a total feature count of $3 \times 4 \times 4 = 48$ per patch, resulting in features of $[H/4, W/4, 48]$. In stage 1, the features are input to the first MMBasicLayer, where they are projected to a specified dimension C through Linear Embedding (implemented by convolution), transforming the features to $[H/4, W/4, C]$. After processing through the Swin Transformer block, stage 1 outputs features of $[H/4, W/4, C]$. Fig. 5 depicts stage 2 of this transformer. The features enter the second MMBasicLayer and undergo patch merging, where four features of size $[H/8, W/8, C]$ are first concatenated along the channel dimension to achieve $[H/8, W/8, 4C]$, and then the channels are reduced from 4C to 2C, resulting in $[H/8, W/8, 2C]$. The features are then imported to this Transformer block, with the output feature scale remaining unchanged. This process is repeated in subsequent stage layers. Finally, a decoder for the specific downstream task is connected to output the required indicators. An example of the framework for Swin Transformer is shown in Fig. 2. Despite these advantages, three main issues arise with visual models during training and application. Large models have excessive activation magnitude differences during training, resolution differences between high-resolution tasks and low-resolution pretraining, and large-scale models require extensive amounts of labeled data for training. To address these issues, the self-supervised learning approach named Swin Transformer V2 has been developed, which can quantify data or train stably to make reliable predictions. This backbone operates on the residual post-norm method and employs Cosine Annealing, which is represented as follows:

$$\text{Sim}(\mathbf{q}_i, \mathbf{k}_j) = \cos(\mathbf{q}_i, \mathbf{k}_j)/\tau + B_{ij} \tag{4}$$

This approach utilizes log-spaced continuous position bias to facilitate the model's transition between different window sizes, solving the migration issue between high and low-resolution tasks. Termed SimMIM, this method reduces the need for labeled data through self-supervised learning, making the training process more efficient [53]. Consequently, we utilize this backbone to address training instability, resolution discrepancies between pretraining and fine-tuning, and the demand for annotated data as shown in Fig. 3.

3.2.2. Plug-in agent attention module stage

Furthermore, we introduce Agent Attention to the above backbones, naming them @LM DeceptionNet-AAST and @LM DeceptionNet-AASTV2. This forms a new quadruple attention paradigm (Q, A, K, V). This module comprises agent attention, bias, and the DWC module [54], which operates in two phases:

$$O = \sigma(QA^T + B_2)\sigma(AK^T + B_1)V + DWC(V), \tag{5}$$

In brief, Agent Attention consists of two operations. The first phase operates on (A, K, V). Agent Tokens A serve as queries to gather information from all value tokens A, K, V into V_A. It processes the information with Query Tokens Q, Key Tokens K, and Value Tokens V, where V is the outcome of the first phase, thereby generating the final output of Agent Attention. In this configuration, Agent Tokens A collect data from K and V and relay it to Q without direct interactions between Q and K, V, enhancing efficiency and reducing redundancy. By pooling Q to form fewer A, Agent Attention minimizes the complexity of Softmax Attention to linear while retaining global modeling capability, balancing efficiency and expressiveness [54]. Building upon these designs, we integrate the Agent Attention module into our visual framework. The approach of pooling original query tokens to create agent tokens has proven to be more effective than other methods. By leveraging the strengths of both Softmax and linear attention, we enhance the computational efficiency and modeling capability of the Swin Transformer V2 model. In practice, we apply the Agent Attention module to the visual model to serve as a new, efficient, robust visual representation learning framework suitable for visual tasks such as deception detection.

3.2.3. Knowledge transfer stage

Simultaneously, to ensure the visual model's generalizability on the deception detection task, we also designed a transfer learning strategy in our proposed models, depicted in Fig. 4. Transfer learning enhances model performance and accelerates the learning process for new tasks by leveraging knowledge from models pre-trained on relevant datasets. During this process, we retained most of the parameter weights from the Real Life Trial and retrained or fine-tuned only the last two layers and the classifier head. This approach prevents overfitting to the new scenarios and preserves the knowledge acquired during pre-training. This stage allowed the new model to adapt to the target task and set the stage for subsequent fine-tuning of the pre-trained network layers.

After the classification layer converged, we unfroze a portion of the pre-trained layers and fine-tuned them at a reduced learning rate to better align their feature representations with the target data distribution, as shown in Fig. 4. By balancing the source and target scenarios, the proposed method maximizes the utilization of pre-trained knowledge while efficiently adapting to the new task and data distributions. In cross-scenario learning, the model must adapt to varying contexts simultaneously, learning shared features across different scenarios. Alternating scenario training helps further by iteratively exposing the model to distinct scenarios, which prevents overfitting to any single domain and gradually enhances the model's adaptability and robustness to feature variability.

In our work, we demonstrated the effectiveness of @LM Deception-Net for visual deception detection tasks, enhancing model performance in visual modalities through the strategic application of transfer learning techniques. Initially, we employed the Real Life Trial dataset, where images underwent standardized preprocessing to ensure consistency and reliability in input data quality. We utilized four variants of our proposed method. To further refine and augment the input data, we applied data augmentation strategies such as Cutout and Mixup, which involved resizing and normalizing the inputs. Specifically, we implemented the Mixup function [72]. It enhances generalization by linearly interpolating between features and labels of training data. This process creates synthetic training samples using the following formula:

$$x = \lambda x_i + (1 - \lambda)x_j \tag{6}$$

$$y = \lambda y_i + (1 - \lambda)y_j \tag{7}$$

where x_i, y_i and x_j, y_j represent pairs of randomly selected data points and their labels from training datasets, x and y are the newly generated samples and labels, and λ is a coefficient sampled from a Beta distribution, Beta(α, α), where α is a hyperparameter that controls the mixing intensity.

Moreover, we explored the SoftTargetCrossEntropy loss function, an adaptation of traditional cross-entropy loss designed to effectively handle soft labels. The formula for this loss function is:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} t_{i,c}\log(p_{i,c}) \tag{8}$$

Here, N denotes sample size per batch, M indicates the total number of categories, $t_{i,c}$ is the soft label value for sample i regarding category c, and $p_{i,c}$ is the probability predicted by the model that sample i belongs to category c, typically derived from the model's softmax output.

We authorized a small, non-zero probability for each classification, which precludes the model from overfitting the training data and boosts its ability to generalize to unseen data. This approach also pushes the model to be more effective at resolving uncertain labels or learning from more complex label configurations, thereby refining the performance and reliability of the final model. The Cross-Entropy loss function utilized in training our visual model is represented as follows:

$$L_v = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i\log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{9}$$

where N represents sample size, y_i denotes the actual label of the i-th sample, either 0 or 1, and $\hat{y}_i$ is the probability predicted by the model that the i-th sample is of the positive class.

As part of the training process, stochastic gradient descent was utilized to minimize the loss function, updating the model parameters by exploiting the gradient computed from a single sample or a small batch at each iteration, thereby expediting computational efficiency. The updated formula is as follows:

$$\theta = \theta - \eta\nabla_\theta J(\theta; x^{(i)}, y^{(i)}) \tag{10}$$

Here, θ represents the model parameters, η the learning rate controlling the step size, and $\nabla_\theta J(\theta; x^{(i)}, y^{(i)})$ the gradient of the loss function computed for the given samples.

After the classifier reached the convergence condition, we unfroze and fine-tuned the last two network layers with a reduced learning rate to better align the feature representations with the target data distribution. The training was iteratively conducted for 30 epochs, with continual monitoring of the loss values to prevent overfitting. Next, the final model was appraised on a reserved test set. The experimental results demonstrated that utilizing a transfer learning strategy enhances the generalizability of deception detection in cross-scenario, validating the effectiveness of our proposed approach.

3.3. Modality fusion strategy generalization

Fig. 6 illustrates our fusion strategy framework for two modalities. We incorporated both visual and auditory modalities to optimize classification performance. Initially, we extracted distinct feature sets: audio features F_{audio} and visual features F_{visual} from their respective models. These features encapsulate crucial information pertinent to their modalities, highlighting potential deceptive intents.

To form a cohesive prediction of deceptive behavior, we explored a modality fusion strategy that captures the information embedded in these features. In this process, F_{audio} and F_{visual} data are processed using Multilayer Perceptrons (MLPs) followed by Softmax layers, which yield preliminary predictions of the audio and visual data Y_{audio} and Y_{visual}. The detailed process is as follows:

Feature Extraction: We employ a late fusion strategy to process audio data with Wav2vec 2.0 and visual data with Swin Transformer and Swin Transformer V2.

$$F_{audio} = \text{Wav2vec2.0}(X_{audio}) \tag{11}$$

$$F_{visual} = \text{Swin Transformer}(X_{visual}) \tag{12}$$

Here, X_{audio} and X_{visual} represent the input streams for the respective modalities.

Modality-Specific Processing: The extracted features are then processed through modality-specific MLPs to generate higher-level representations suitable for classification.

Audio MLP Output:

$$H_{audio} = \sigma(W_{audio} \cdot F_{audio} + b_{audio}) \tag{13}$$

$$Z_{audio} = \sigma(W^o_{audio} \cdot H_{audio} + b^o_{audio}) \tag{14}$$

Visual MLP Output:

$$H_{visual} = \sigma(W_{visual} \cdot F_{visual} + b_{visual}) \tag{15}$$

$$Z_{visual} = \sigma(W^o_{visual} \cdot H_{visual} + b^o_{visual}) \tag{16}$$

where σ is the activation function (e.g., ReLU or Sigmoid), W_{audio}, W_{visual} and $W^o_{audio}, W^o_{visual}$ are the weight matrices for the hidden and output layers, respectively, and b_{audio}, b_{visual}, and $b^o_{audio}, b^o_{visual}$ are the biases.

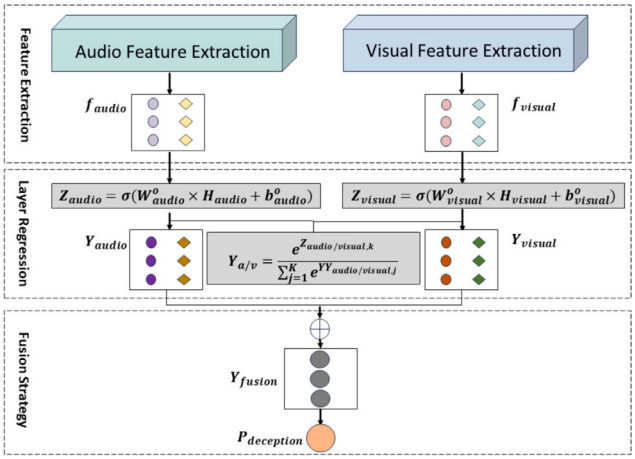


Fig. 6. Fusion Strategy Framework: An advanced framework for fusion method of audio and visual features.

The outputs Z_{audio} and Z_{visual} from the audio and visual models are then sent to a Softmax function to predict deception classification for each modality.

For the audio modality:

$$Y_{audio} = \left(\frac{e^{Z_{audio,k}}}{\sum_{j=1}^{K} e^{Z_{audio,j}}} \right) \tag{17}$$

where $Z_{audio,k}$ denotes the score of class k derived from audio features, and K is the number of classes, with j iterating over all classes.

Similarly, for the visual modality:

$$Y_{visual} = \left(\frac{e^{Z_{visual,k}}}{\sum_{j=1}^{K} e^{Z_{visual,j}}} \right) \tag{18}$$

where $Z_{visual,k}$ represents the score for class k derived from visual features.

This formulation crucially allows the model to interpret the linear scores as class probabilities, an essential feature for the task of deception classification. By implementing Softmax separately for each modality, we clarify how linear scores are transformed into a probabilistic distribution, enhancing our understanding of how the model processes and integrates information from different modalities to make a final decision.

Feature Fusion: The outputs Z_{audio} and Z_{visual} are integrated using a weighted linear combination:

$$Y_{fusion} = \beta Y_{audio} + (1 - \beta) Y_{visual} \tag{19}$$

where β is a weight parameter balancing the contributions from each modality.

Final Prediction: This fused output is processed through a layer of Softmax to determine the final multimodal deception score:

$$P_{deception} = \left(\frac{e^{Y_{fusion,k}}}{\sum_{j=1}^{K} e^{Y_{fusion,j}}} \right) \tag{20}$$

where K is the number of classes.

Within this fusion framework, the distinct dominance of audio and visual data is integrated, thereby minimizing the latent weaknesses inherent in single-modality systems. As a result, it enhances the accuracy and robustness of the deception detection task.

4. Experiment

This section details a series of experiments conducted using benchmark datasets to validate the feasibility of our proposed multimodal prediction approach and transfer learning strategy. Initially, we outline the preprocessing steps for the deception detection data, followed by a detailed tabulation of the experimental setup, and finally, we detail the evaluation metrics used to appraise the proposed methods.

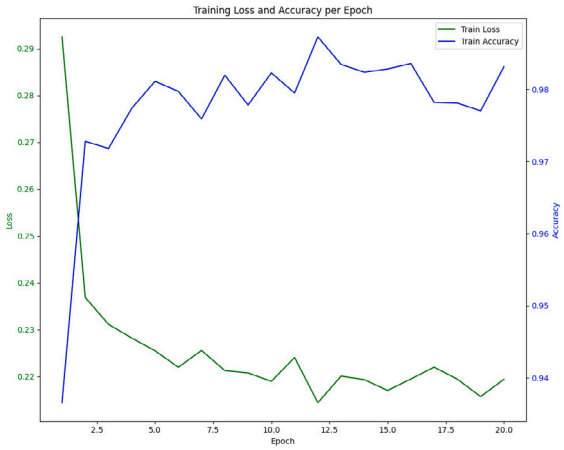
4.1. Datasets and baselines

We assess the proposed method using the Real Life Trial dataset [18], Box of Lies dataset [24], and DOLOS dataset [16], which are commonly used for unscripted deception detection scenarios. Each instance in our three datasets contains ground-truth annotations and multimodal cues, such as text, audio, and visual. We project these multimodal cues onto a deception cues layout following their standard practice used in deception detection.

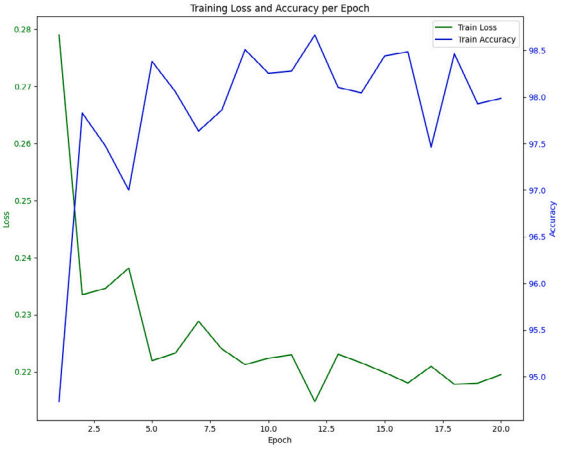
Real Life Trial: The Real Life Trial dataset consists of 121 video segments from criminal court trials in the contemporary United States, featuring 56 subjects. These videos include 60 truthful and 61 deceptive clips [18]. After filtering out several corrupted videos, we obtained 110 videos (55 truthful and 55 deceptive) involving 53 subjects, which were used for our experiments.

Box of Lies: Box of Lies comprises conversations from The Tonight Show Starring Jimmy Fallon's Box of Lies game, where participants describe objects to deceive their opponents, who must determine whether the description is truthful. The dataset includes multimodal annotations of verbal and non-verbal behaviors, such as facial expressions and linguistic features, to facilitate deception detection in dialogues. Initial classification experiments demonstrate that these multimodal features achieve accuracy levels surpassing both random guessing and human baselines, reaching up to 69% [24].

DOLOS: The DOLOS dataset consists of publicly available materials from a British reality comedy show on YouTube and involves speakers describing personal life-related statements, which the audience must judge for veracity. Each dialogue round is recorded and labeled according to the accuracy of the speaker's description [16]. We downloaded a total of 1678 videos from YouTube using the DOLOS dataset annotation file, slightly more than the original count of 1675. Compared to Box of Lies [24], it covers much broader and more diverse image content and provides three types of protocols for evaluating the performance of deception detection: the Train–Test Protocol, Duration Protocol, and Gender Protocol. The Train–Test Protocol uses three-fold average cross-validation to assess deception detectors. The Duration Protocol addresses variations in duration in real-world scenarios, categorizing clips as either short (2–4 s) or long (5–10 s) and cross-using them during training and testing. The Gender Protocol explores the impact of gender on deception detection by dividing the dataset into male and female

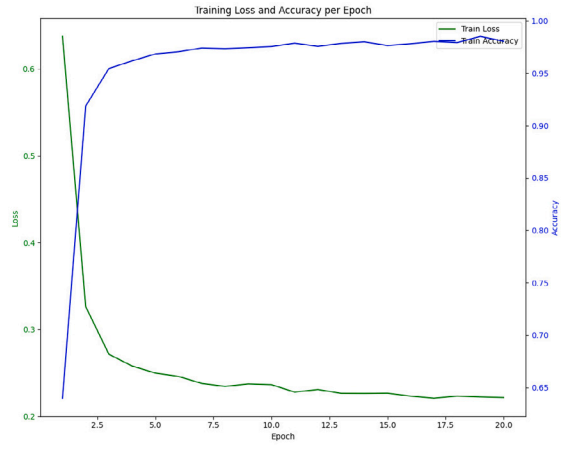


(a) A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-ST.

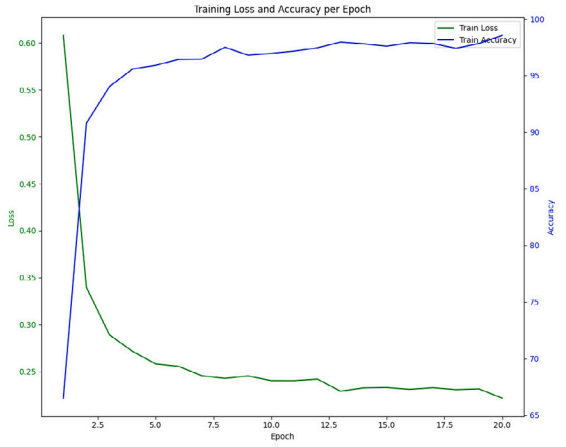


(b) A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-STV2.

Fig. 7. A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-ST and @LM DeceptionNet-STV2.



(a) A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-AAST.



(b) A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-AASTV2.

Fig. 8. A pre-trained accuracy-loss curve of 20 Epochs in the Real Life Trail dataset employed @LM DeceptionNet-AAST and @LM DeceptionNet-AASTV2.

clips, which are then cross-utilized for training and testing. These protocols serve as baselines to explore the influence of factors such as gender and duration on deception detection performance, enhancing model generalizability and providing a comprehensive evaluation framework.

Preprocessing. The @LM DeceptionNet layout representation used in training and testing covers three scenarios. For the Real Life Trial, there are 121 videos. We extract video clips, output them in spectrogram and image format, and filter out corrupted content accordingly. For Box of Lies, there are 25 videos, but without satisfactory quality. The frequent viewpoint changes cause mismatching between the visual cues and the audio cues in their clips. To ensure a fair comparison, we preprocessed and filtered the dataset. For DOLOS, clips of raw videos related to deceptive content were first excerpted based on their annotations. The audio tracks from these clips were then processed to enable the W2V2 feature extractor to output as spectrograms, each represented in a 64×512 feature dimension. Given the model input size requirements, facial images were resized to 224×224 for the @LM DeceptionNet-ST and @LM DeceptionNet-AAST and 256×256 for the @LM DeceptionNet-STV2 and @LM DeceptionNet-AASTV2.

In the audio module, features were extracted using the first four transformer encoder layers of W2V2, as deception detection tasks benefit from low-level features [5,37,73]. To enhance the visual module's performance, we adopted data augmentation techniques, including the cutout technique for resizing and normalizing data, along with defining the Mixup function. Additionally, the Agent Attention module was integrated into the visual model to reduce computational complexity and capture detailed visual information. The pre-trained parameter weights were frozen, and only the last two layers and output classification layer were retrained and partially fine-tuned on the Box of Lies dataset and DOLOS dataset to prevent overfitting and potential loss of trained knowledge.

4.2. Training details

We have three stages of our training. In the first stage, the audio features were extracted via the first four encoder layers of Wav2vec2.0, and the model was trained on the Real Life Trial dataset, Box of Lies dataset, and DOLOS datasets to adapt to the audio features. As described above, the visual features are trained with aggressive augmentation consisting of resizes, rotations, normalization, color jitters, label smoothing, and switch probability.

Transformers. In the first phase, we implement transformers to facilitate training. The Adam optimizer and AdamW optimizer were adopted

Table 2
Abbreviations used in the results and discussion section.

Abbreviation	Description
RLT	Real Life Trial Dataset
ST	Swin Transformer
STV2	Swin Transformer V2
AAST	Agent Attention Based Swin Transformer
AASTV2	Agent Attention Based Swin Transformer V2
A + V	Audio + Visual

with a learning rate of 1×10^{-4}, dynamically adjusted through a cosine annealing strategy. Model training was conducted on a computing platform with Nvidia A100 GPUs, with a batch size of 32. We trained for 100 epochs, evaluating performance on their test datasets. In the second phase, we trained for 20 epochs on the Real Life Trial dataset, which also had the same learning rate. After each epoch, model performance was evaluated based on accuracy and loss, as seen in Figs. 7 and 8.

In the transfer learning phase, we employed pre-trained models on the Real Life Trial dataset to fine-tune and retrain these models, using a learning rate of 1×10^{-5} for 30 epochs on the Box of Lies dataset and the DOLOS datasets. After completing the training of these new layers, the last three layers of the previously trained model were unfrozen and fine-tuned to enhance performance on the Box of Lies dataset and the DOLOS dataset. This approach exemplifies the transfer learning strategy effectively and optimizes training efficiency.

Additionally, we introduce a lightweight attention mechanism into the visual models as a variant to enhance computational performance. We project this attention mechanism in the visual models to train on the three datasets with reduced parameters compared to the original models.

4.3. Evaluation metrics

In deception detection, the primary objective is to accurately identify deceptive behavior using computational models. Our experiments involve both audio and visual features, and we evaluate their performance using Accuracy (ACC) and F1 scores. To assess computational efficiency, we report the number of floating-point operations (FLOPs) and the number of parameters, as presented in Table 3. These metrics are calculated during the evaluation phase by processing images through four variants of our proposed @LM DeceptionNet with a batch size of 32, representing the average computational cost per inference.

5. Results and discussion

In this section, we evaluate the results for each modality. Through a series of ablation experiments, we highlight the benefits of applying the transfer learning strategy to new scenarios. Finally, we conduct a comparative analysis of our approach with state-of-the-art deception detection techniques on the DOLOS dataset. Table 2 lists all the abbreviations used in the forthcoming analysis.

5.1. Quantitative result

5.1.1. Statistical significant test

We employ the Mann–Whitney U test to compare the performance of visual models between the full training and transfer learning phases. Specifically, Figs. 9 and 10 present box plots illustrating the accuracy distributions of the four model variants—ST, STV2, AAST, and AASTV2—on the Box of Lies and DOLOS datasets, supplemented with statistical test results to ensure the reliability of our conclusions.

We show the differences of the Mann–Whitney U test between the full training against transfer learning phases in different datasets in Fig. 9. Notably, the performance of ST and STV2 exhibits eminent variations on the Box of Lies dataset. By applying non-parametric tests such as

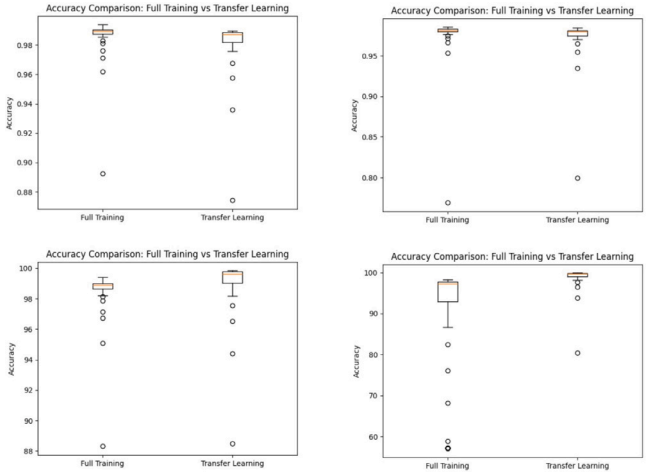


Fig. 9. Box Plot of Accuracy: Full Training vs Transfer Learning with Mann–Whitney U Test Results. From top left to bottom right they are ST on the Box of Lies dataset, ST on the DOLOS dataset, STV2 on the Box of Lies dataset, and STV2 on the DOLOS dataset.

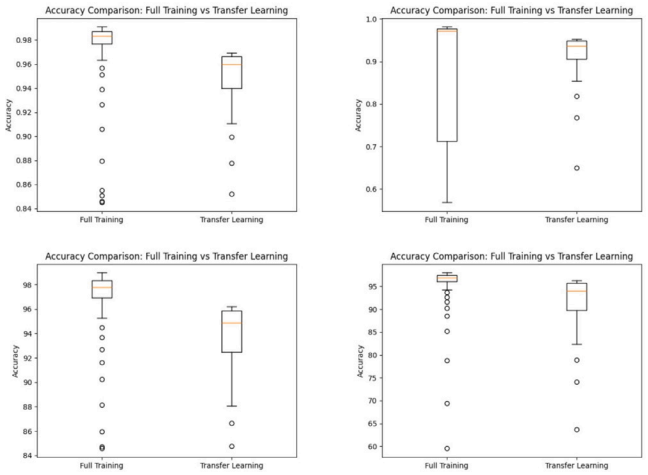


Fig. 10. Box Plot of Accuracy: Full Training vs Transfer Learning with Mann–Whitney U Test Results. From top left to bottom right they are AAST on the Box of Lies dataset, AAST on the DOLOS dataset, AASTV2 on the Box of Lies dataset, and AASTV2 on the DOLOS dataset.

the Mann–Whitney U test to evaluate these differences, the resulting p-values indicate that the performance of some models during the transfer learning phase differs significantly from that during the full training phase.

We repeat the experiments from above using AAST and AASTV2 in Fig. 10. The Mann–Whitney U test results also demonstrate the models' performance during the transfer learning phase is eminently different from during the full training phase. This is particularly eminent in the DOLOS dataset, where the performance differences of the AAST against AASTV2 models are more pronounced.

5.1.2. Computation complexity

Table 3 summarizes the training parameters and computational complexities of various @LM DeceptionNet model variants during both full training and transfer learning phases. Significant differences in computational requirements emerge among these models. In the full training phase, we trained four model variants from scratch: ST, STV2, AAST, and AASTV2. Each exhibited distinct computational complexities and parameter counts. The ST model, based on the Swin Transformer architecture, had the highest computational complexity with 8.56×10^9

Table 3
Summary of training parameters and computational complexity for @LM DeceptionNet in different phases. The first part represents the full training phase, while the second part corresponds to the transfer learning phase.

Method	Backbone	Input resolution	Epochs	Params	FLOPs	Pre-trained
Full training stage						
ST	swin_small_patch4_window7_224	224 × 224	100	48.84M	8.56×10^9	ImageNet-1K
STV2	swinv2_tiny_window8_256	256 × 256	100	27.58M	4.38×10^9	ImageNet-1K
AAST	swin_small_patch4_window7_224	224 × 224	100	27.52M	4.38×10^9	ImageNet-1K
AASTV2	swinv2_tiny_window8_256	256 × 256	100	28.42M	5.75×10^9	ImageNet-1K
Transfer learning-stage methods						
ST	swin_small_patch4_window7_224	224 × 224	20	48.84M	8.56×10^9	ImageNet-1K
STV2	swinv2_tiny_window8_256	256 × 256	20	27.58M	4.38×10^9	ImageNet-1K
AAST	swin_small_patch4_window7_224	224 × 224	20	27.52M	4.38×10^9	ImageNet-1K
AASTV2	swinv2_tiny_window8_256	256 × 256	20	28.42M	5.75×10^9	ImageNet-1K
ST	swin_small_patch4_window7_224	224 × 224	30	47.64M	8.56×10^9	Real Life Trail
STV2	swinv2_tiny_window8_256	256 × 256	30	26.37M	4.38×10^9	Real Life Trail
AAST	swin_small_patch4_window7_224	224 × 224	30	26.03M	4.38×10^9	Real Life Trail
AASTV2	swinv2_tiny_window8_256	256 × 256	30	26.81M	5.75×10^9	Real Life Trail

Table 4
Audio and visual deception detection on the Real Life Trail (RLT) dataset and Box of Lies dataset. The metrics are ACC (%) and F1 (%).

Modality	Dataset	Method	Epoch	ACC	F1
Audio	RLT	Wav2vec2.0	100	**81.93**	79.74
	Box of Lies			72.58	**78.91**
Visual	Real Life Trail	ST	100	99.95	99.94
		STV2		99.96	99.93
		AAST		**99.98**	99.98
		AASTV2		99.87	99.86
	Box of Lies	ST		99.91	99.85
		STV2		**99.95**	99.85
		AAST		99.94	99.79
		AASTV2		99.88	99.62
Visual-Transfer Learning	RLT-Box of Lies	ST	20+30	99.79	99.32
		STV2		**99.91**	99.70
		AAST		99.17	97.33
		AASTV2		97.76	92.82

FLOPs and 48.84 million parameters, reflecting substantial resource needs. In contrast, STV2 and AAST required 4.38×10^9 FLOPs and had approximately 27.5 million parameters, indicating lower computational demands. AASTV2, with 5.75×10^9 FLOPs and 28.42 million parameters, was more complex than STV2 and AAST but more efficient than ST.

During the transfer learning phase, all models underwent fewer training epochs (20 or 30) using pre-trained weights from the Real Life Trial dataset. While the number of parameters remained nearly constant, reduced epochs decreased training time and computational requirements. For instance, ST retained 47.64 million parameters and the same FLOPs (8.56×10^9), indicating that transfer learning fine-tuned existing weights rather than retraining from scratch. Similarly, STV2, AAST, and AASTV2 maintained their parameter counts and FLOPs, preserving computational efficiency during transfer learning. Our analysis also highlights the impact of input resolution on computational complexity. ST and AAST used an input resolution of 224 × 224, while STV2 and AASTV2 employed a higher resolution of 256 × 256. The increased resolution led to higher FLOPs, particularly for AASTV2, which had 5.75×10^9 FLOPs, exceeding AAST's complexity. This indicates that higher resolutions may enhance model accuracy but require more computational resources.

As such, ST is the most computationally intensive model, suitable for environments with abundant computational resources. Conversely, STV2, AAST, and AASTV2, with lower FLOPs and parameter counts, are more appropriate for resource-constrained settings. Transfer learning significantly reduced training costs by limiting epochs while maintaining parameter integrity, making it advantageous for deployment in environments with limited training capabilities. Moreover, carefully selecting input resolution enables a balance between accuracy and computational efficiency, emphasizing the trade-offs in our model design.

5.2. Performance of audio and visual deception detection

As seen in Table 4, the audio model based on Wav2vec2.0 showed varying performance between the two datasets. This audio model achieved an accuracy of 81.93% and an F1 score of 79.74% on the Real Life Trial dataset, indicating a balanced detection of deceptive and truthful instances. On the Box of Lies dataset, the accuracy was 72.58%, while the F1 score was 78.91%, presenting a better balance between precision and recall.

The visual models ST, STV2, AAST, and AASTV2 demonstrated exceptional performance on both datasets. On the RLT dataset, ST achieved an accuracy of 99.95% and an F1 score of 99.94%. Meanwhile, STV2, AAST, and AASTV2 all achieved accuracies exceeding 99.8% and F1 scores near 99.9%, reflecting high reliability in deception detection. On the Box of Lies dataset, these models again surpassed 99.8% accuracy, with F1 scores ranging from 99.62% (AASTV2) to 99.94% (AAST), indicating the effectiveness of visual features with minimal differences among the models.

In the transfer learning phase, where models were trained on the RLT dataset and tested on the Box of Lies dataset, ST maintained high performance with 99.79% accuracy and an F1 score of 99.32%, while STV2 achieved 99.19% accuracy and a 97.90% F1 score. However, AAST and AASTV2 experienced larger declines, with AASTV2 showing the lowest performance at 97.76% accuracy and a 92.82% F1 score. This illustrates that AAST and AASTV2 are more sensitive to domain shifts compared to ST and STV2, which retained high accuracy and F1 scores.

Table 5 presents the accuracy and F1 scores of audio and visual methods for deception detection on the DOLOS dataset. It tabulates audio, visual, and visual-transfer learning with three types of protocols. We adopted Wav2vec2.0 trained for 100 epochs and achieved an

Table 5
Audio-Visual deception detection on DOLOS dataset. The metrics are ACC (%) and F1 (%).

Modality	Dataset	Method	Epoch	3-fold average		Duration protocol		Gender protocol	
				ACC	F1	ACC	F1	ACC	F1
Audio	DOLOS	Wav2vec2.0	100	59.35	**66.74**	56.02	59.78	52.61	62.83
Visual	DOLOS	ST	100	79.29	75.21	78.05	72.89	46.75	25.70
		STV2		**80.02**	75.68	77.29	72.61	46.03	31.46
		AAST		78.23	74.07	77.65	72.78	45.77	34.31
		AASTV2		78.93	74.89	77.23	71.82	43.85	33.49
Visual-Transfer Learning	RLT-DOLOS	ST	20+30	78.44	74.25	78.39	73.07	52.09	33.88
		STV2		78.64	73.94	**78.77**	73.52	44.67	30.44
		AAST		77.03	72.69	78.14	70.96	44.18	35.70
		AASTV2		76.48	70.06	74.43	65.18	48.91	27.37

Table 6
Performance of the fusion method in the Real Life Trial dataset, the Box of Lies dataset, and the DOLOS dataset. The metric is ACC (%).

Modality	Dataset	ACC (%)
A+V-ST	Real Life Trial	**94.53**
A+V-STV2		93.71
A+V-AAST		92.18
A+V-AASTV2		91.08
A+V-ST	Box of Lies	85.42
A+V-STV2		85.16
A+V-AAST		83.25
A+V-AASTV2		**87.24**
A+V-ST	DOLOS	74.70
A+V-STV2		**75.88**
A+V-AAST		74.27
A+V-AASTV2		74.62

Table 7
The comparison of our approach with state-of-the-art method in the DOLOS dataset.The metric is ACC (%).

Modality	Methods	ACC (%)
Audio	MLP-MFCC [4]	56.86
	MLP-OpenSMILE [5,74,75]	57.49
	DOLOS Audio [16]	59.19
	Ours	**59.35**
Visual	LSTM-Action Unit [5,42]	57.47
	LSTM-Facial Affect [5,40]	57.67
	LSTM-OpenFace [5,37]	56.81
	LSTM+RN18[3–5,37,40,41]	59.55
	MLP-MUMIN [18]	58.84
	ViT-Face [16]	61.44
	Ours	**80.02**
A + V	Score Fusion-OpenSMILE + Face [5,37,75]	60.00
	Score Fusion-OpenSMILE + OpenFace [5,40]	59.75
	DOLOS Concat [3,40,41]	61.62
	PAVF [16]	64.75
	Ours	**75.88**

average accuracy of 59.35% and an F1 score of 66.74% in the 3-Fold Average protocol. In the duration protocol, the accuracy dropped to 56.02% and the F1 score to 59.78%. In the gender protocol, accuracy decreased to 52.61%, with an F1 score of 62.83%.

In the visual modality trained on DOLOS for 100 epochs, the variant STV2 attains the highest 3-fold average accuracy of 80.02% and an F1 score of 75.68%, making it the most effective visual model in this context. ST closely follows with an accuracy of 79.29% and an F1 score of 75.21%. AAST and AASTV2 achieve slightly lower accuracies of 78.23% and 78.93%, respectively, with F1 scores around 74%–75%. In the duration protocol, STV2 also performed well, while AAST showed a better precision–recall balance with a higher F1 score of 72.78%. However, all visual models experience significant drops in accuracy and F1 scores in the gender protocol, with accuracies ranging from 43.85% to 46.75% and F1 scores between 25.70% and 35%.

In the transfer learning phase, the visual model trained on the RLT dataset and tested on DOLOS demonstrates good generalization. The variant of ST achieves an accuracy of 78.44% and an F1 score of 74.25%. STV2 and AAST have similar accuracies of 78.64% and 77.03%, respectively, with STV2 achieving the highest duration protocol accuracy of 78.77%. AAST performs well under the duration protocol, with an accuracy of 78.14% and an F1 score of 70.96%. AASTV2 shows slightly lower performance with an accuracy of 76.48% and an F1 score of 70.06%. Despite this, all models perform poorly under the gender protocol, with accuracies ranging from 44.18% to 52.09% and low F1 scores, highlighting significant challenges in generalizing across different gender groups.

5.3. Performance of fusion strategy

In Table 6, we tabulate the performance of the fusion method across three datasets: Real Life Trial, Box of Lies, and DOLOS. In the Real Life Trial dataset, the A+V-ST achieves the highest accuracy at 94.53%, closely followed by A+V-STV2 at 93.71%. The A+V-AAST and A+V-AASTV2 display accuracies of 92.18% and 91.08%, respectively. For the Box of Lies dataset, A+V-AASTV2 attains the best accuracy at 87.24%, outperforming A+V-ST at 85.42% and A+V-STV2 at 85.16%. The A+V-AAST records an accuracy of 83.25%. Furthermore, in the DOLOS dataset, A+V-STV2 achieved the highest accuracy at 75.88%, followed by A+V-ST at 74.70%, A+V-AAST at 74.27%, and A+V-AASTV2 at 74.62%.

5.4. Comparison with state-of-the-art methods

To further demonstrate the effectiveness of our proposed methods, Table 7 compares the ACC scores of our method against state-of-the-art methods in audio, visual, and audio-visual fusion modalities. The proposed audio model achieved an ACC of 59.35%, which is 2.49% higher than the 56.86% achieved by MLP-MFCC and 1.86% higher than the 57.49% achieved by MLP-Open SMILE.

We next show the comparison of the proposed model and its variants against state-of-the-art methods on the DOLOS testset in Table 7. In comparing with the state-of-the-art approaches, @LM DeceptionNet-STV2 surpasses LSTM-Action Unit, LSTM-Facial Affect, LSTM-Open Face and LSTM+RN18, by 22.55%, 22.35%, 23.21%, and 20.47% with 100 epochs. We also show that @LM DeceptionNet-STV yields a 21.18% and 18.58% gain in ACC on the DOLOS over MLP-MUMIN and ViT-Face.

Our multimodal fusion approach A+V achieves state-of-the-art performance on the DOLOS by exceeding the Score Fusion-Open SMILE + Face, Score Fusion-Open SMILE + Open Face, Concat, PAVF by 15.88%, 16.13%, 14.26% and 11.13%, respectively. These comparative results demonstrate the efficacy of our fusion strategy in multimodal fusion approaches and detection frameworks.

5.5. Ablation analysis

Table 8 displays an ablation analysis of a visual model with ST, STV2, AAST, and AASTV2 as four variants. For relative comparison,

Table 8
The ablation analysis in metric ACC (%) was achieved by cross-scenario without transfer learning.

	Method	Train Data-RLT		Train Data-Box of Lies		Train Data-DOLOS	
		Test Box of Lies	Test DOLOS	Test RLT	Test DOLOS	Test RLT	Test Box of Lies
V	ST	31.69	47.55	56.45	56.44	54.47	63.81
	STV2	26.46	49.14	56.45	57.03	54.54	62.01
	AAST	**60.55**	46.81	56.29	**57.68**	**62.66**	**64.70**
	AASTV2	40.07	**50.76**	**56.47**	56.97	50.30	60.50

we use Real Life Trial (RLT), Box of Lies, and DOLOS. We began by training the models on RLT and testing them on Box of Lies. AAST achieves an accuracy of 60.55%, surpassing ST at 31.69%, STV2 at 26.46%, and AASTV2 at 40.07% by 28.86%, 34.09% and 20.48%, respectively. Conversely, when testing on DOLOS, AASTV2 shows an accuracy of 50.76%, over STV2 at 49.14% and ST at 47.55%, while AAST lagged at 46.81%. When training on DOLOS and testing on RLT, it attains 62.66% accuracy (vs. STV2: 54.54%, Δ8.12; ST: 54.47%, Δ8.19), while Box of Lies testing yielded 64.70% accuracy (ST: 63.81%, Δ0.89; STV2: 62.01%, Δ2.69). Notably, training protocols from Box of Lies to RLT show marginal inter-variant differences (56.29%–56.47% range), suggesting architectural distinctions primarily enhance cross-domain adaptation rather than scripted-to-controlled detection. Hence, it is apparent that our proposed framework properly addresses the generalizability and shortens the complexity of deception scenarios for improved detection performance and efficacy. This ablation analysis provides a foundation for better comprehending each model's ability to handle data in new scenarios.

6. Applications

Detecting deceptive behavior has many applications. In this section, we explore the applications of detection for deception in real-life scenarios.

Since one motivation for our task definition lies in the generalizability of the deception network, we wish to determine whether this enables our model to apply in new scenes from out-of-scenario. We generalize to new scenarios using a knowledge-learning approach, as shown in Table 4 and Table 5. Most importantly, our model can be applied to real-world settings such as border security, criminal investigations, and psychological research [76]. For instance, similar applications like iBorderCtrl [8] are used in border security projects to analyze non-verbal behavior and distinguish between truth and deception, assessing travelers' identities, intentions, or possessions. In criminal investigations, a deception network can also be employed for deception detection of suspects. Compared to large-scale screening, this structured and controlled application is more feasible [77]. As a psychological research tool, our model can serve as a foundational tool for studying the cognitive mechanisms of deception. These applications exhibit the potential of deception detection, particularly in the aforementioned fields.

7. Conclusion

In this paper, we propose a multi-modality approach for deception detection that integrates feature learning across multiple modalities, such as audio and vision. The @LM DeceptionNet model introduces an Agent Attention mechanism applied to the visual transformer model, enhancing deception detection performance in the visual modality. Additionally, it employs a multimodal feature fusion approach, effectively combining audio and visual models to improve the accuracy of deception detection.

In addition, we outline a transfer learning strategy to transition training results from the source dataset to the target dataset, thereby enhancing the generalization capability of visual models across various datasets. Experimental results on different datasets demonstrate the effectiveness of the plug-in Agent Attention-based visual model, the multimodal fusion method, and the transfer learning strategy. Apart from this, we explored critical technologies such as multimodal feature representation, fusion, and knowledge transfer in deception detection tasks, providing robust support for the development of a deception detection system.

Building on the initial successes and insights obtained from our current research, our future work will focus on refining and expanding the generalization capabilities of our modality fusion strategies. This effort will be particularly concentrated on enhancing model fusion techniques, a critical component for advancing the domain applicability of multimodal deception detection systems.

CRediT authorship contribution statement

Yuanya Zhuo: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis. **Vishnu Monn Baskaran:** Writing – review & editing, Writing – original draft, Validation, Supervision. **Lillian Yee-Kiaw Wang:** Writing – review & editing, Validation, Supervision. **Raphaël C.-W. Phan:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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